

Scaling Effects of Neuronal Population Size on ECoG-Based Face and House Decoding

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2025-05-12

Background

- Previous studies (Miller et al, 2016) have shown that it is possible to predict the type of stimuli (face, house, or other) using the broadband activity from face-selective and house-selective cells with up to 96% accuracy (of shown visual stimuli) with a less than 5% false positive rate and a ~20ms error in timing.
- Responses of a neuronal population, measured through electrocorticography (ECoG), can be used to predict and decode the perceptual state of a patient with relatively high accuracy.
- Shifts in broadband activity (wide frequency range) usually reflect local neuronal population's firing rate and represent a generic correlate of local cortical function across a range of brain areas and behavioral tasks.

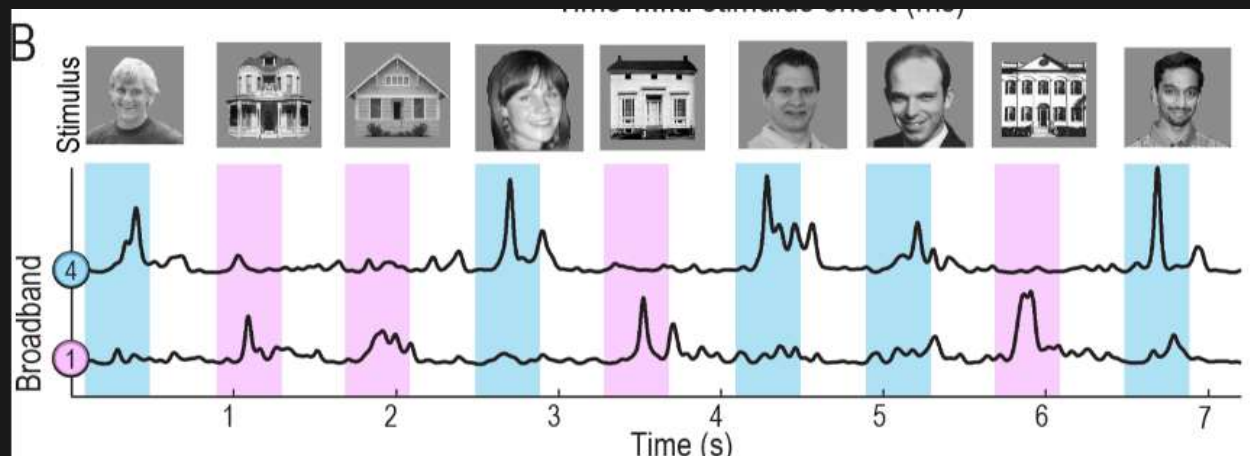
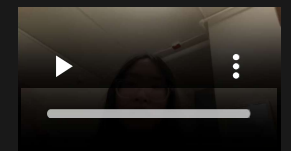


Figure 1. From Miller et al, 2016. Task: Grayscale faces and houses were displayed in random order for 400ms each, with 400ms of a blank gray screen between each picture. Subjects were asked to report a simple target. From the raw potential, the time course of broadband spectral change was extracted from each brain site where blue = faces; pink = houses.



Background (CONT.)

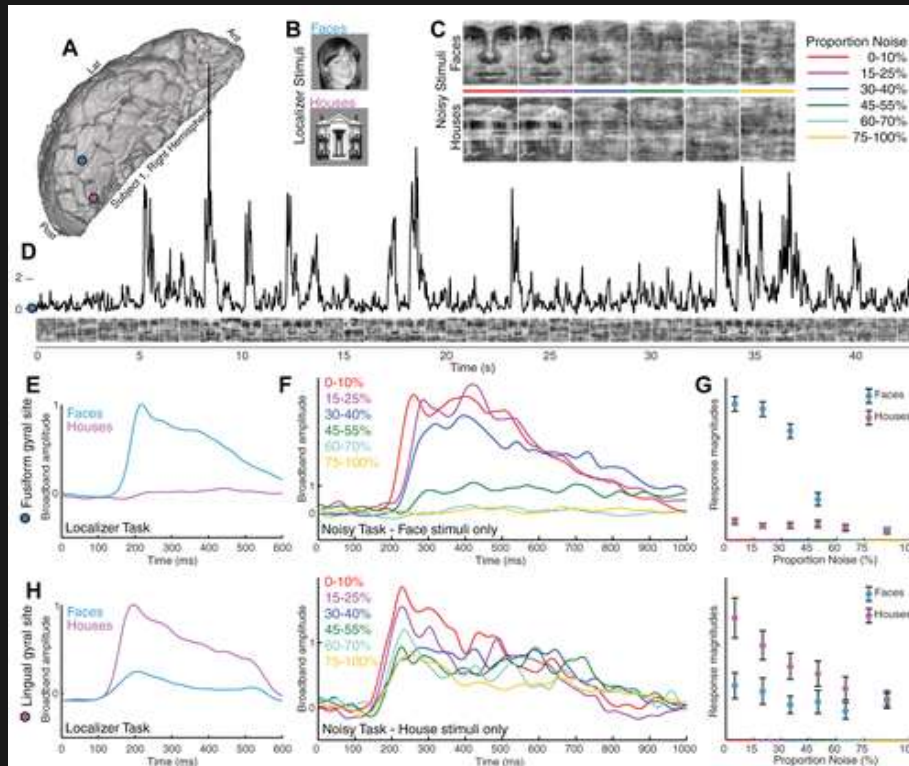
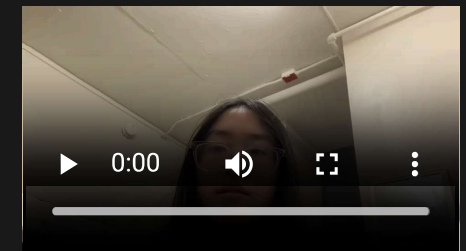


Figure 2. From Miller et al, 2017. A) shows ECoG electrode locations B) basic task: pictures of grayscale faces and houses were displayed in random order for 400 ms each, with 400-ms blank screen in between. C: noisy task. Phase-scrambled close-up pictures of faces and houses were shown for 1 s each. D: broadband spectral change in the electrical potential. E: averaged broadband response templates to face and house stimuli (from A) generated from the localizer task. F: averaged broadband responses to face stimuli (as noise increases). G: the face template, generated from the localizer task, is projected to single trials from the noisy task, revealing a robust neurometric function (mean $\pm$ SE). H: same as E–G but for the lingual gyral (pink) site in A, with the use of a house template to quantify single-trial responses (far right)

- VTC (ventral temporal cortex) utilize both visual properties and categorical (post-perceptual) recognition to form face percept
- Posterior fusiform gyrus (face-selective): neural activity would decrease proportionally as visual noise increased. However, once passing a threshold (too much noise), the activity levels returned to baseline.



Background (CONT.)

- Regions in the Ventral Temporal Cortex that show face and object selectivity and spatiotopical organization.
- Fusiform face area (FFA) has shown face-selectivity based on holistic processing. Larger responses to faces than other objects, especially in the ventral occipitotemporal cortex (VOTC), and peaks in FFA.

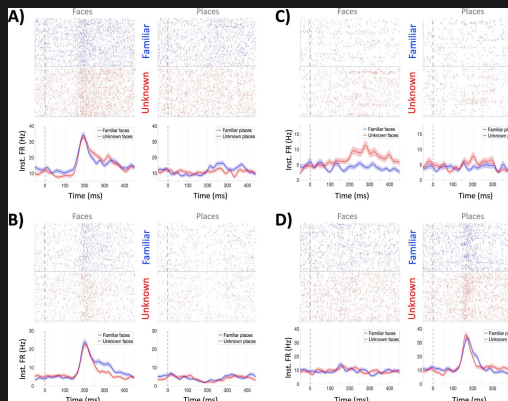
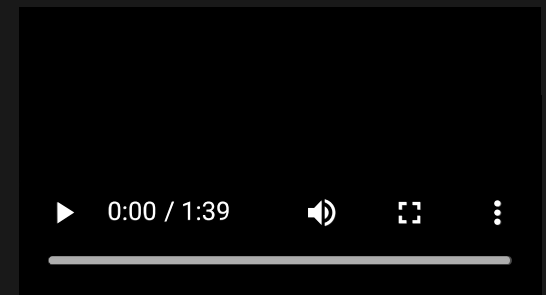


Figure 3. Different single neurons responding to pictures of faces (and in general, not to pictures of places), with varying responses to familiar and unknown faces, using a two-sided t-test. Figures B,C are $p < 0.05$. A shows a single neuron responding to pictures of faces, B shows preference to familiar faces, C shows preference to unknown faces, and D shows a multi-unit with a significant response to places.



Main Question / Hypothesis

- Will decreasing the size of the neuronal population (resolution) result in a decrease in accuracy of identifying faces and houses? At what rate does this decline occur?
- We predict that accuracy will decrease as population size of neurons (resolution) decreases. Reducing scale will make the identifications based off of broadband activity more inaccurate.



About the Dataset

- Seven patients (four male, three female) with ECoG implants, over two sessions, sampled at a rate of 1000Hz.
- Shown grayscale face and house images (contrast and luminance balanced)
- ECoG: electrical activity from the brain is recorded through subdural electrodes implanted inside the cranium
- high spatial (mm scale) and temporal (ms scale) resolution
- Implanted for epilepsy patients prior to surgical treatment
- Compared to electroencephalography (EEG): ECoG has higher spatial and temporal resolution, which is sometimes necessary to locate epileptic zones

General Approach

- Replicate results seen in previously mentioned study
 - identify face-selective and house-selective neurons
 - Average broadband activity for each respective population of neurons when shown an image of face or house
- Scale down
 - Some neurons have stronger responses
 - Randomly select which neuron is dropped
 - Drop by one channel, then two, etc
- Run twice to see if which neurons are chosen affects rate of decrease
 - randomly selected channels vs decrease numerically
 - compare

Methods - Data and Preprocessing

- ECoG datasets were collected using a visual categorization task with face and house stimuli
- Broadband signals were accessed from a public dataset, “faceshouses.npz”
- Preprocessing involved a two-stage filtering of the raw voltage signals from each electrode
 1. High-pass filter at 50Hz to isolate broadband activity
 2. The analytic signal was then squared to compute instantaneous broadband power
 3. Low-pass filter at 10Hz to smooth the power estimate



Methods - Epoch Extraction

- Epochs are short, time-aligned, segments of neural data. In this case, it is centered around the onset of each visual stimulus.
- To isolate these epochs, broadband power signals were separated around each stimulus onset time using a window of 600ms spanning from -200ms to +400ms.
- Trials were grouped into two categories based on the stimulus type
 1. Face Stimuli - $\text{stim_id} > 50$
 2. House Stimuli - $\text{stim_id} \leq 50$



Methods - Electrode-Level Analysis

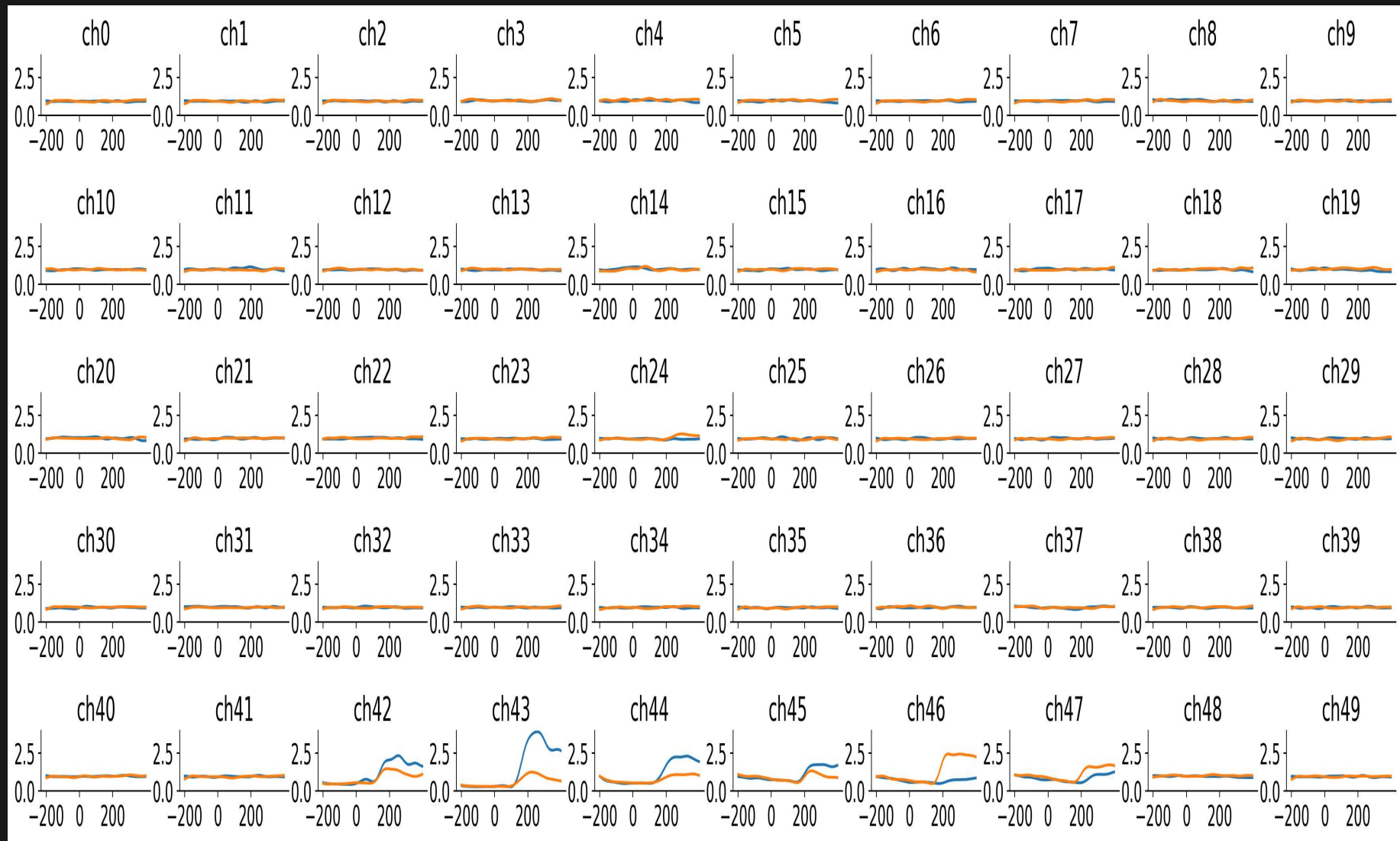
- CODE FROM all_electrodes, see electrodes to determine which distinguish faces and which distinguish houses, and trials_one_electrode.
1. Identification of which electrodes show reliable differential activity between faces vs. houses
 - For each of the 50 electrodes, indexed by j, function all_electrodes creates a subplot in a 5x10 grid layout.
 - For each channel, it plots mean power over time for house trials ($V_{\text{house}}[:,j]$) and face trials ($V_{\text{face}}[:,j]$)
 - X-axis time range is centered around stimulus onset, and Y-axis limits are fixed to [0,4] to standardize all the electrodes
 2. Visualizing Trial-by-Trial Dynamics at a Single Electrode via Heatmapping
 - Extract trial-by-time matrices for each electrode (baseline-corrected and z-scored per trial), sort by condition (faces vs. houses), and plot as a time-aligned heatmap (−200 to +800 ms) with a unified color scale to compare response magnitudes.
 - Overlay markers for stimulus onset and condition boundaries to visualize consistency of category-specific power modulations across individual trials.



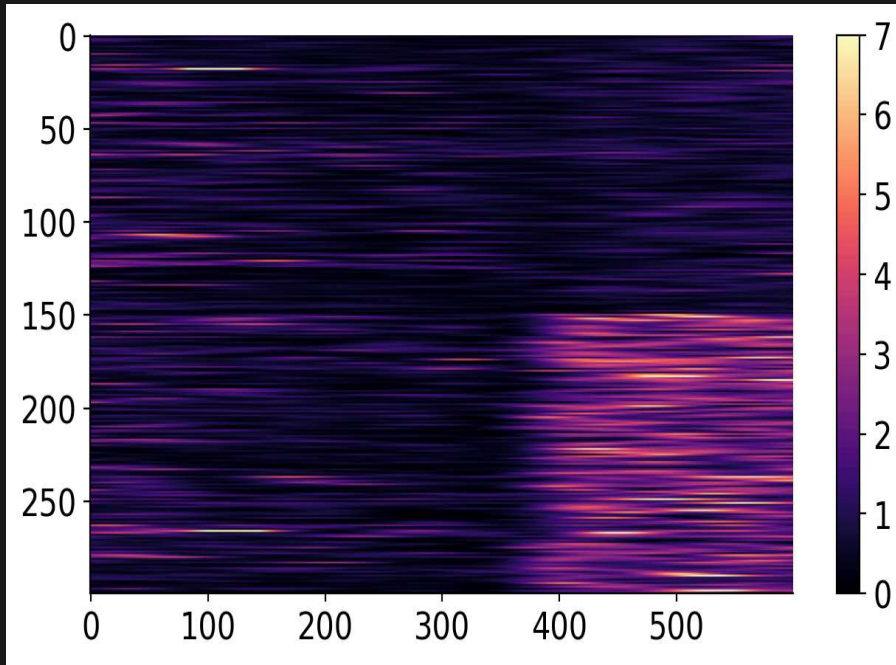
Results Video



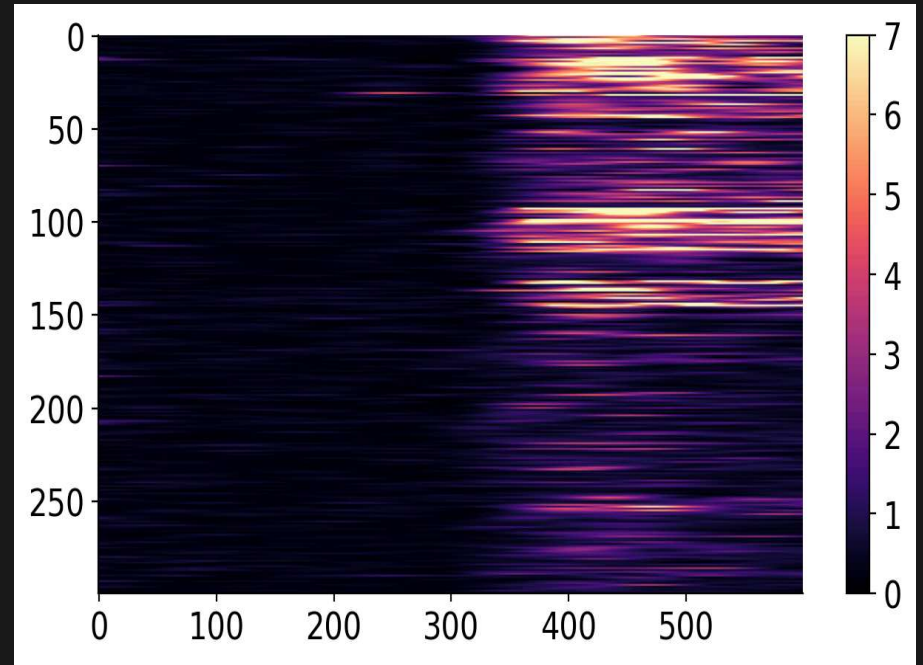
Results (1)



Results (2)

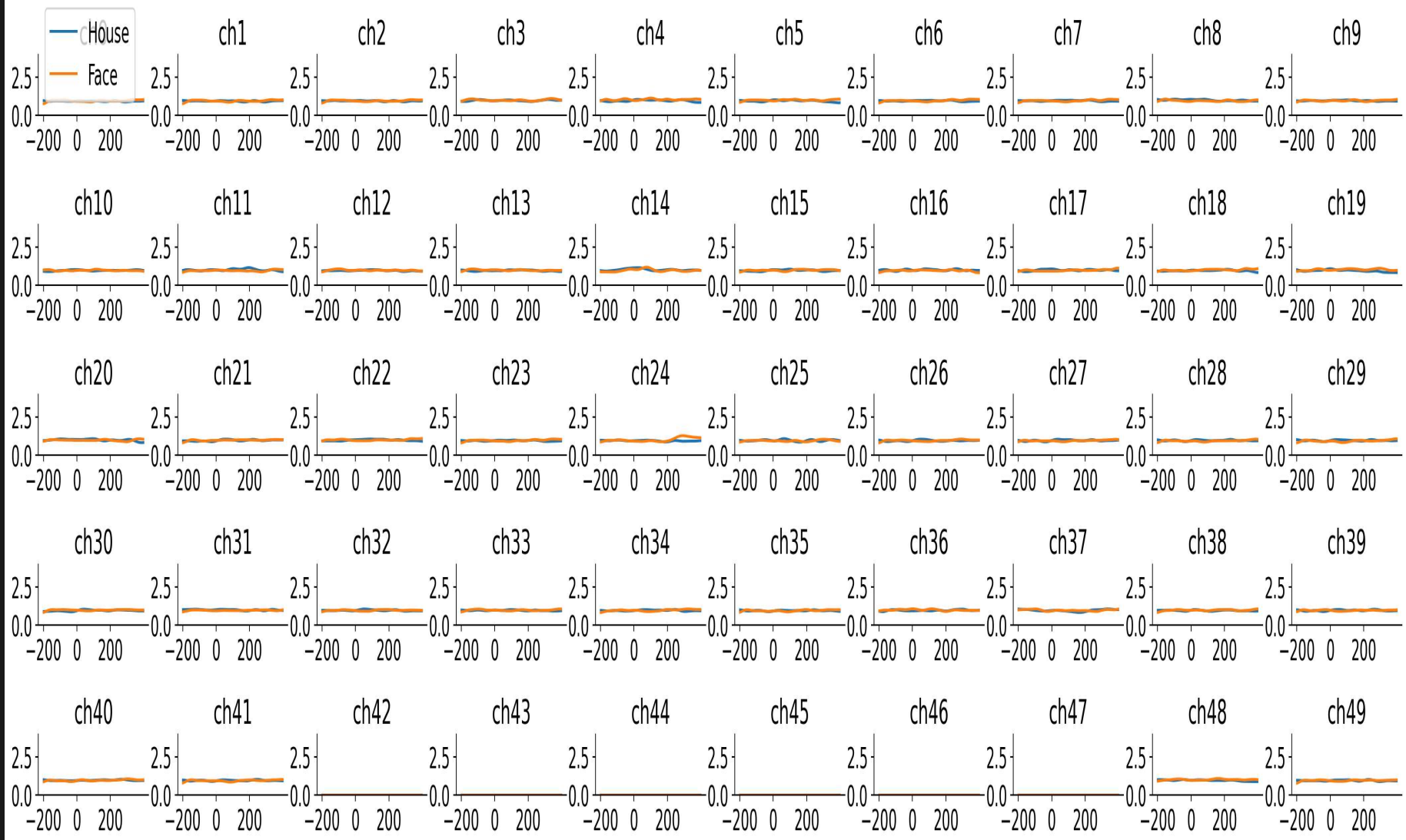


- Electrode 46 is shown to respond strongly to faces. This graph demonstrates the responses for all the face trials in electrode 46.



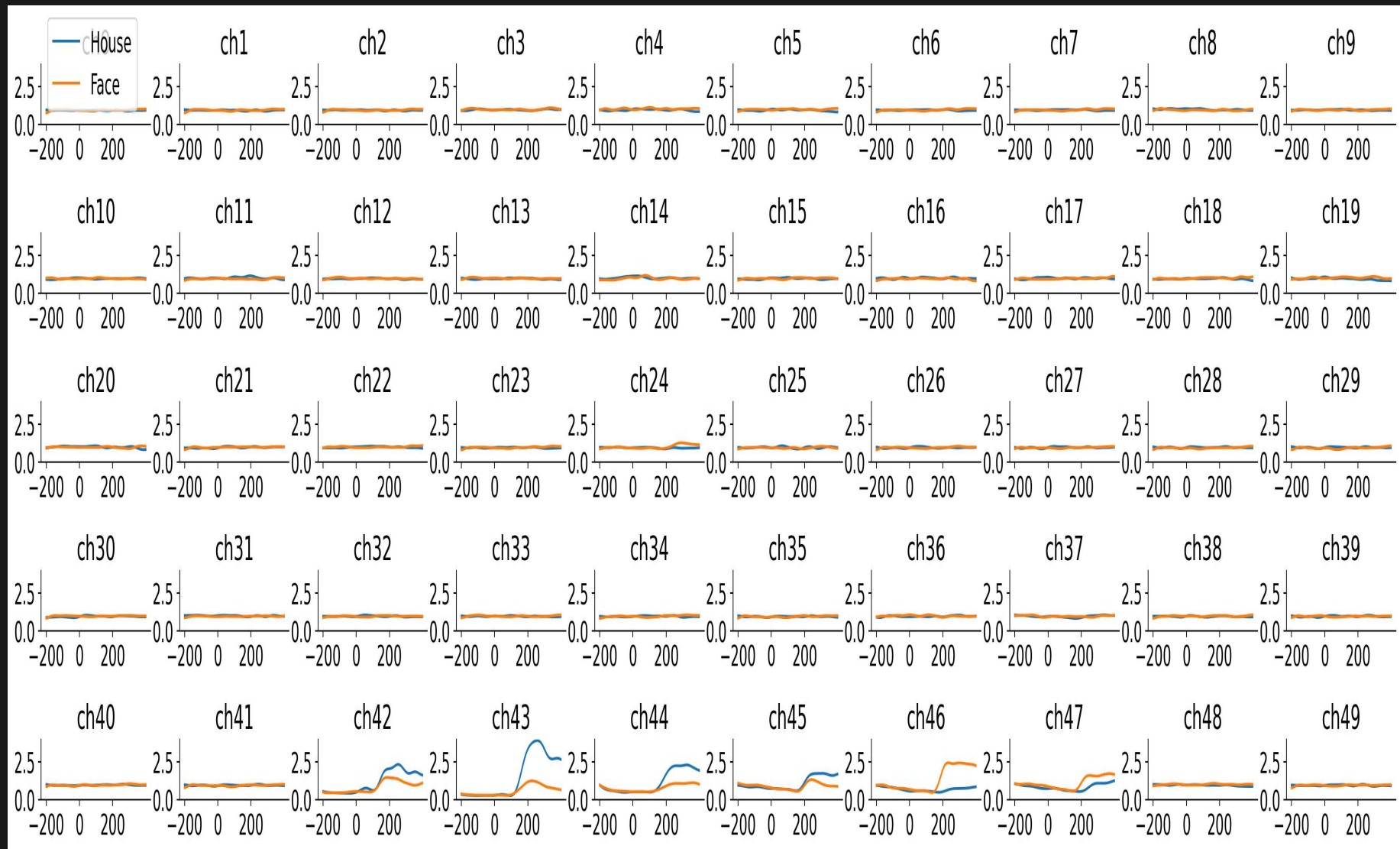
- Electrode 43 is shown to respond strongly to faces. This graph demonstrates the responses for all the house trials in electrode 43.

Results (3)



- Channels 42-47 was lesioned / zeroed out

Results (4)



Results (5)

Discussion

- Lesioned channels demonstrate that the strength of response decreases dramatically
- Unable to predict face or house when channels are lesioned
- However, we ran into issues then applying our lesioned channels to overall data and to understand to what degree data was affected (quantitatively)
- Misunderstanding of the applications of dataset: we thought all broadband activity from all channels was averaged together in the heatmaps, only to realize they identified electrodes that responded to houses (a few channels)

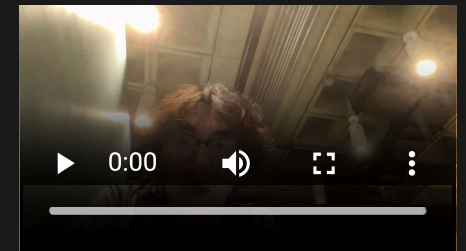


Conclusion

- Lesioning channels decreases broadband activity and would decrease the accuracy of predicting house or face stimuli
- Experimental design needs improvement; better understanding of the dataset and how to apply

Future Directions:

- Follow through with computing the exact numbers of new, decreased accuracy
- How to lesion data without rendering it uninterpretable



Resources

1. [Spontaneous Decoding of the Timing and Content of Human Object Perception from Cortical Surface Recordings Reveals Complementary Information in the Event-Related Potential and Broadband Spectral Change](#)
2. [Face Percept Formation in Human Ventral Temporal Cortex](#)
3. [The Physiology of Perception in Human Temporal Lobe Is Specialized for Contextual Novelty](#)
4. [Neuromatch ECoG Datasets](#)
5. [Electrocorticography \(ECoG\)](#)
6. [Epilepsy: Electrocorticography](#)
7. [Single neuron responses underlying face recognition in the human midfusiform face-selective cortex](#)
8. [Face inversion reveals holistic processing of peripheral faces](#)